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自进化多智能体系统

从 system-level 和 agent-level 实现共同进化



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MAS²: SELF-GENERATIVE, SELF-CONFIGURING, SELF-RECTIFYING MULTI-AGENT SYSTEMS

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**把构建MAS这个任务本身交给多智能体系统来完成
让一个 meta-MAS 自动生成、配置并修正 target MAS**

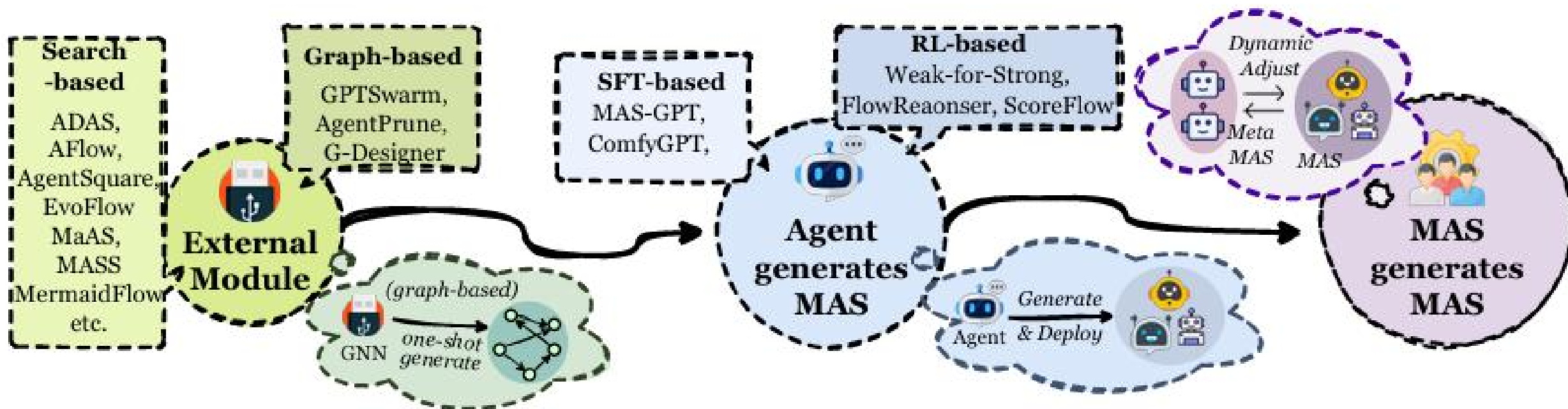
Motivation

manual design -> full automation

The core problem of automatic multi-agent system: **generate-once-and-deploy**

真实环境的任务执行过程中经常会暴露出生成阶段看不到的问题。

MAS 不应该只是在开始时被一次性生成，而应该能在执行过程中根据反馈继续修正



Methods

形式化MAS:

$$M = \langle R, P, T, B \rangle$$

角色设计 + 通信协议 + 工具选择 + 模型分配

三个 meta-agent:

Generator: $M_{temp} = \langle R, P, T \rangle$

Implementor: $\phi : R \rightarrow L$

Rectifier:

$$A_R(s_t) = 1[C(s_t) > \theta_C \vee O(s_t) = Failure]$$

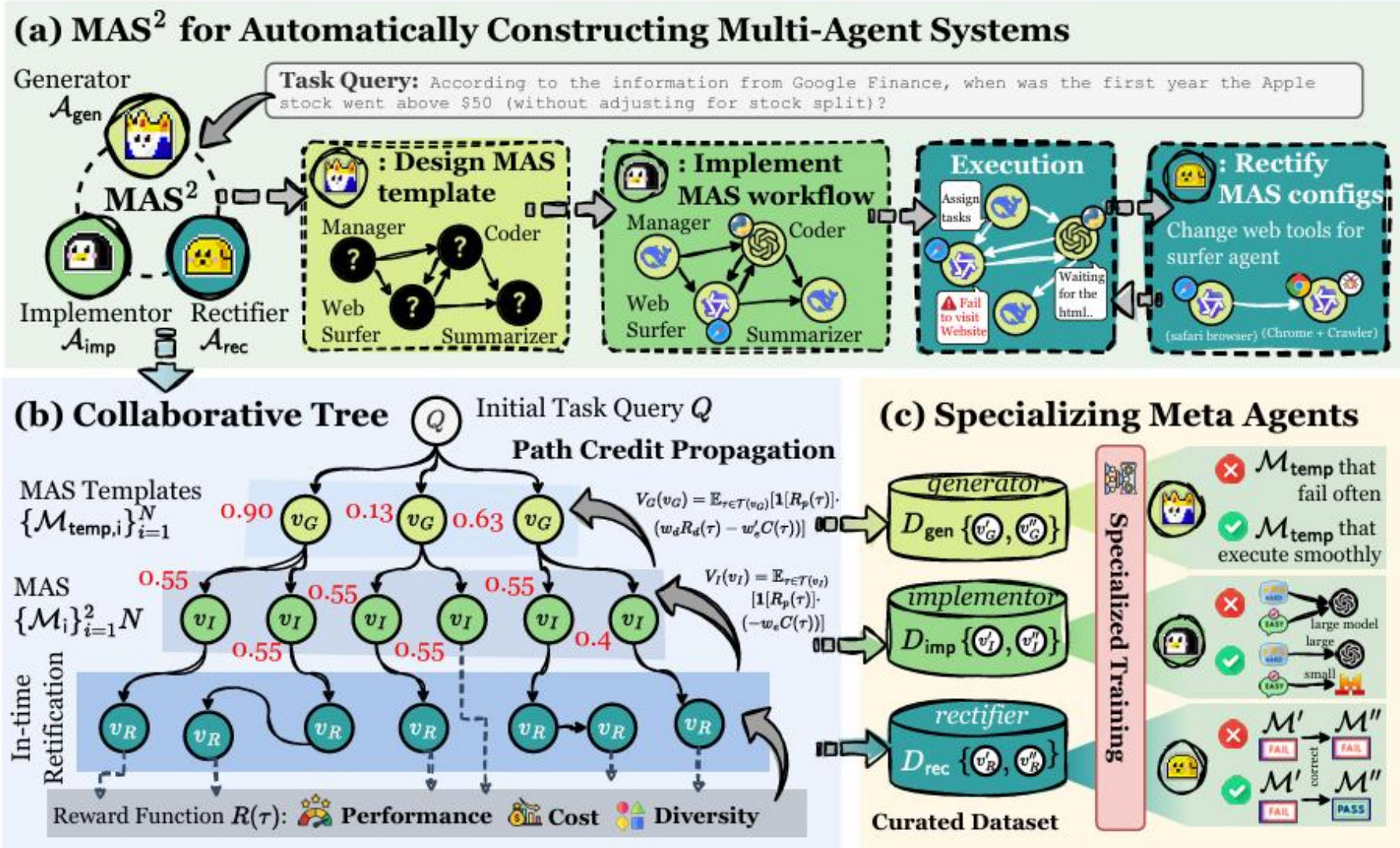


Figure 2: The framework overview of our proposed MAS².

Methods

CTO: Collaborative Tree Optimization

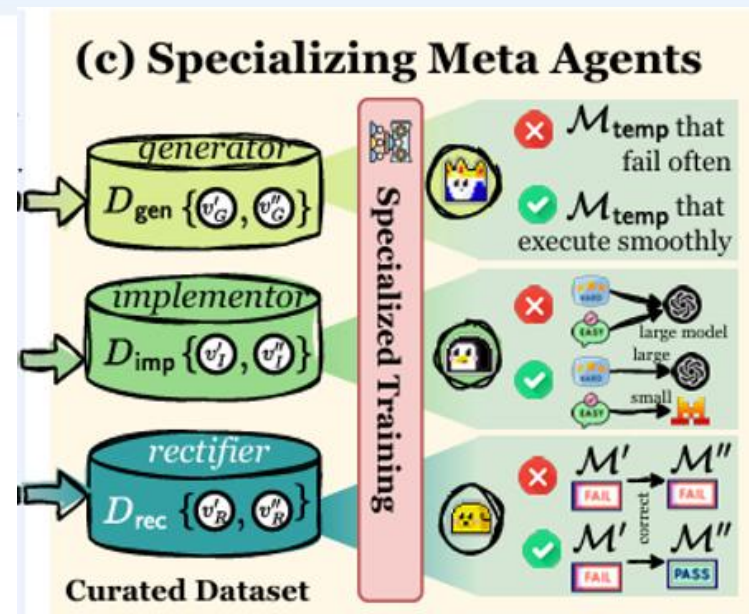
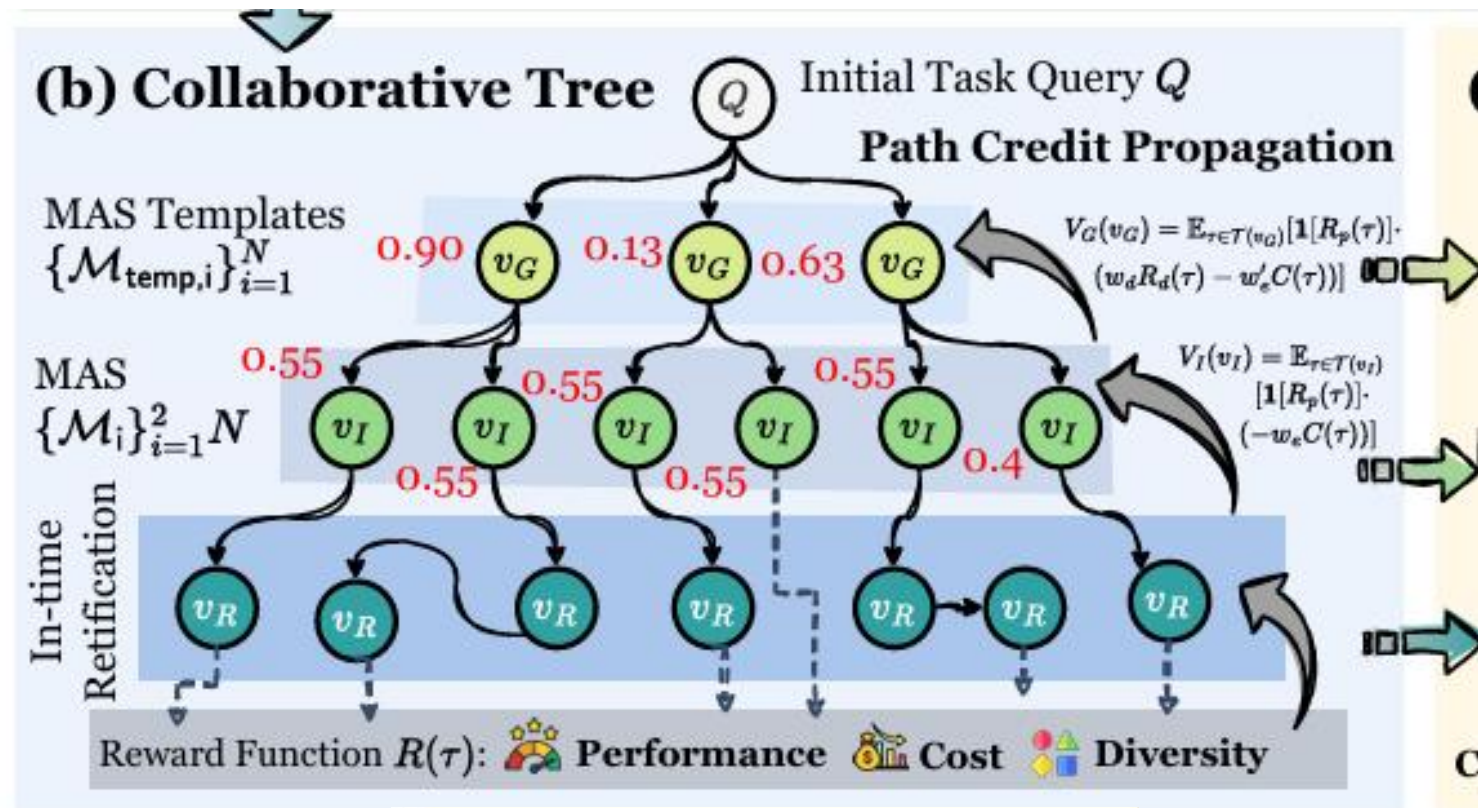
1.多种MAS设计 构建collaborative tree

2.给每条路径基于结果和成本给 reward

3.反向分配给中间决策节点
中间节点的 value = 所有经过这个节点的路径 reward 的平均值

4.构造 偏好数据
根据 Δvalue , 训练时权重越大

得到Dgen、Dimp、Drec三种数据
分别训练三个 meta-agent
倾向选择高value的系统设计、模型配置和修正动作



Case Study

Benchmark	任务需求	MAS ² 设计
Bamboogle	多跳问答, 答案验证	多候选答案 + 强模型判断
HumanEval	代码生成, bug 修复	并行 code generation + test, 强模型审查整合
BrowseComp+	深度研究, 信息检索复杂	规划 + 搜索 + 文档召回 + 摘要 + 反思

Case Study Figure 5 shows task-specific multi-agent systems designed by MAS² that strategically allocate different LLMs, each with customized complexity and workflow configurations.

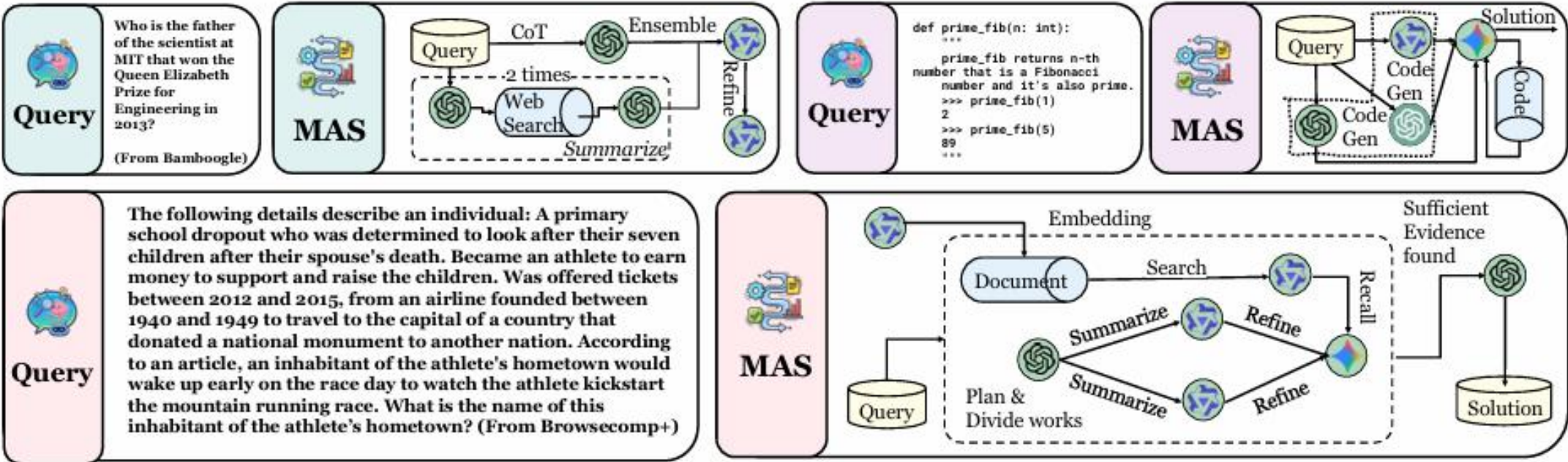


Table 1: **Performance comparison** across 8 benchmarks across 13 baselines. Each cell reports the average of **three random runs**. The best/second-best results are **bolded/underlined**.

Model	Multi-hop Search			Deep Research	Code Generation		Math
	HotpotQA	Bamboogle	NQ	BrowseComp+	HumanEval	MBPP	MATH
Qwen3-14B	62.2	34.4	51.9	13.1	75.6	40.8	61.5
GPT-4o-mini	64.1	26.4	70.7	13.7	87.8	46.0	51.2
QwQ-32B	64.9	39.2	61.5	2.8	51.8	82.2	65.6
Qwen-2.5-72B	66.8	31.2	63.0	2.6	82.3	76.0	55.3
GPT-4o	69.5	49.6	71.1	13.2	89.6	73.4	56.5
Avg. (Above)	65.5	36.2	63.6	9.5	77.4	63.7	58.0
CoT (GPT-4o)	66.2 $\uparrow 0.7$	52.8 $\uparrow 16.6$	68.7 $\uparrow 5.1$	<u>16.1$\uparrow 7.6$</u>	90.8 $\uparrow 13.4$	75.4 $\uparrow 11.7$	58.9 $\uparrow 0.9$
SC (GPT-4o)	66.4 $\uparrow 0.9$	54.4 $\uparrow 18.2$	73.6 $\uparrow 10.0$	11.8 $\uparrow 2.3$	<u>96.3$\uparrow 18.9$</u>	75.1 $\uparrow 11.4$	62.0 $\uparrow 4.0$
MedPrompt	72.4 $\uparrow 6.9$	48.0 $\uparrow 11.8$	66.1 $\uparrow 2.5$	12.0 $\uparrow 2.5$	92.1 $\uparrow 14.7$	69.2 $\uparrow 5.5$	59.7 $\uparrow 1.7$
MultiPersona	71.1 $\uparrow 5.6$	50.2 $\uparrow 14.0$	68.4 $\uparrow 4.8$	10.1 $\uparrow 0.6$	92.9 $\uparrow 15.5$	70.4 $\uparrow 6.7$	53.7 $\downarrow 4.3$
LLM-Debate	66.9 $\uparrow 1.4$	54.4 $\uparrow 18.2$	70.8 $\uparrow 7.2$	15.3 $\uparrow 5.8$	88.7 $\uparrow 11.3$	70.3 $\uparrow 6.6$	67.3 $\uparrow 9.3$
DyLAN	80.8 $\uparrow 15.3$	59.7 $\uparrow 23.5$	72.1 $\uparrow 8.5$	15.8 $\uparrow 6.3$	90.4 $\uparrow 13.0$	77.3 $\uparrow 13.6$	65.7 $\uparrow 7.7$
ADAS	78.5 $\uparrow 13.0$	50.8 $\uparrow 14.6$	65.9 $\uparrow 2.3$	7.0 $\downarrow 2.5$	88.8 $\uparrow 11.4$	68.7 $\uparrow 5.0$	51.7 $\downarrow 6.3$
MaAS	83.6 $\uparrow 18.1$	62.0 $\uparrow 25.8$	76.0 $\uparrow 12.4$	14.0 $\uparrow 4.5$	92.8 $\uparrow 15.4$	82.2 $\uparrow 18.5$	70.1 $\uparrow 12.1$
AFlow	77.9 $\uparrow 12.4$	59.2 $\uparrow 23.0$	74.5 $\uparrow 10.9$	10.0 $\uparrow 0.5$	92.9 $\uparrow 15.5$	82.9 $\uparrow 19.2$	68.5 $\uparrow 10.5$
ScoreFlow	<u>86.0$\uparrow 20.5$</u>	<u>64.8$\uparrow 28.6$</u>	<u>76.4$\uparrow 12.8$</u>	10.4 $\uparrow 0.9$	95.9 $\uparrow 18.5$	<u>84.7$\uparrow 21.0$</u>	64.4 $\uparrow 6.4$
MAS ²	89.3$\uparrow 23.8$	67.2$\uparrow 31.0$	79.1$\uparrow 15.5$	19.7$\uparrow 10.2$	97.0$\uparrow 19.6$	85.1$\uparrow 21.4$	71.3$\uparrow 13.3$

在四类任务上评估

baseline 分三类
单模型 / prompting
手工设计 MAS
自动化设计MAS

1. 已有 MAS 方法泛化不稳定
2. MAS² 在多数任务上取得较好的结果

在推理时加入了训练时未见过的强模型

MAS² 的 meta-agent 能够把新模型纳入系统配置中，合理利用不同 backbone 的能力，也得到性能的提升

关键：如何根据任务和角色把不同 backbone 放到合适位置

Table 2: Performance results when integrated with stronger unseen LLMs. “Vanilla LLM” denotes merely using one LLM backbone for testing. Checkmarks are used to denote the LLMs used.

Method	Stronger LLMs			Datasets			
	Qwen3-Coder	GPT-5-Mini	Gemini-2.5-Pro	MATH		Bamboogle	
				Perf.(%)	Cost(\$)	Perf.(%)	Cost(\$)
Vanilla LLM	✓			69.7	0.54	32.8	0.06
			✓	68.0	14.48	83.2	2.27
		✓		87.3	1.22	80.8	0.21
MAS ²	✗	✗	✗	71.3	0.74	67.2	0.15
	✓	✓	✗	88.6	2.22	82.5	0.53
	✓	✗	✓	84.8	15.47	84.2	2.57
	✗	✓	✓	90.6	16.14	84.0	2.73

Experiments

cost-performance trade-off 分析

MAS² 处在新的 Pareto frontier 上:
它比相近成本的方法更准, 也比相近准确率的方法更便宜

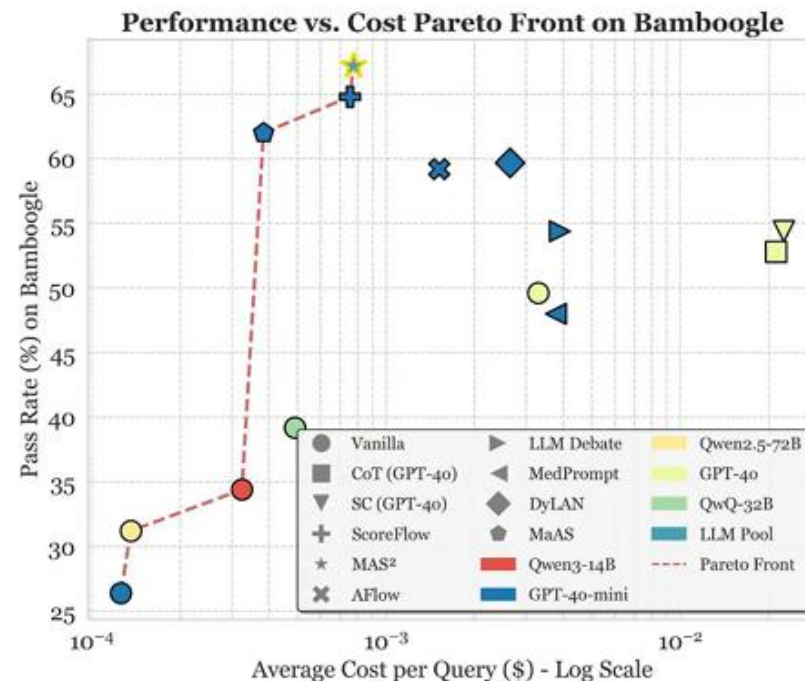
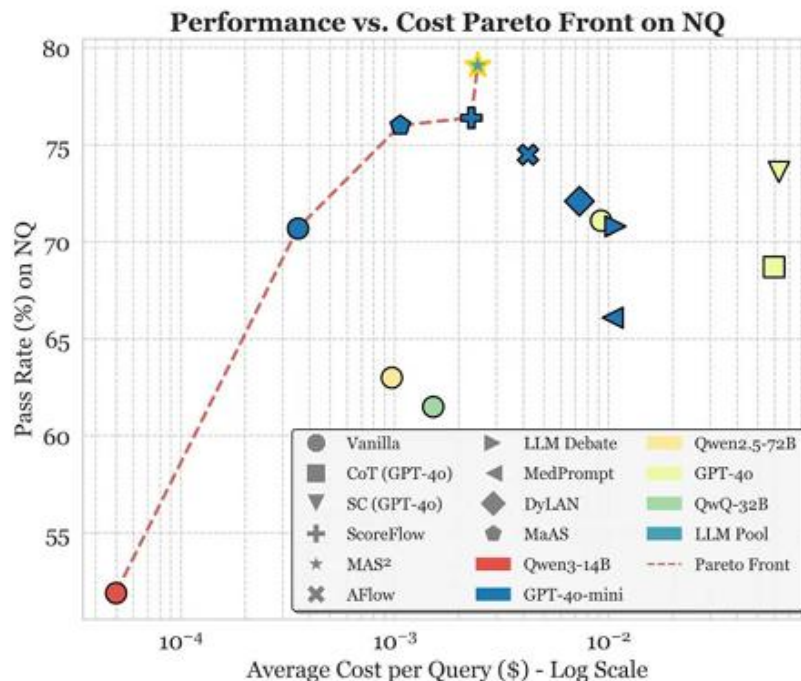


Figure 3: Cost-performance trade-off on the NQ and Bamboogle benchmark.



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CoMAS: Co-Evolving Multi-Agent Systems via Interaction Rewards

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多个 agent对同一个问题轮流提出解法、批评、打分
将分数转化为reward来实现不断提升和进化

Motivation

使用强化学习训练 agent 时,
reward 从哪里来

External rewards

通常依赖 verifier 或 reward model
在可验证任务上有效, 但不够通用

Intrinsic rewards

减少了对外部标注的依赖, 但不一定可靠

Interaction rewards

来源于人类学习中的讨论、批评、协作
将多agent之间的interaction作为
reward

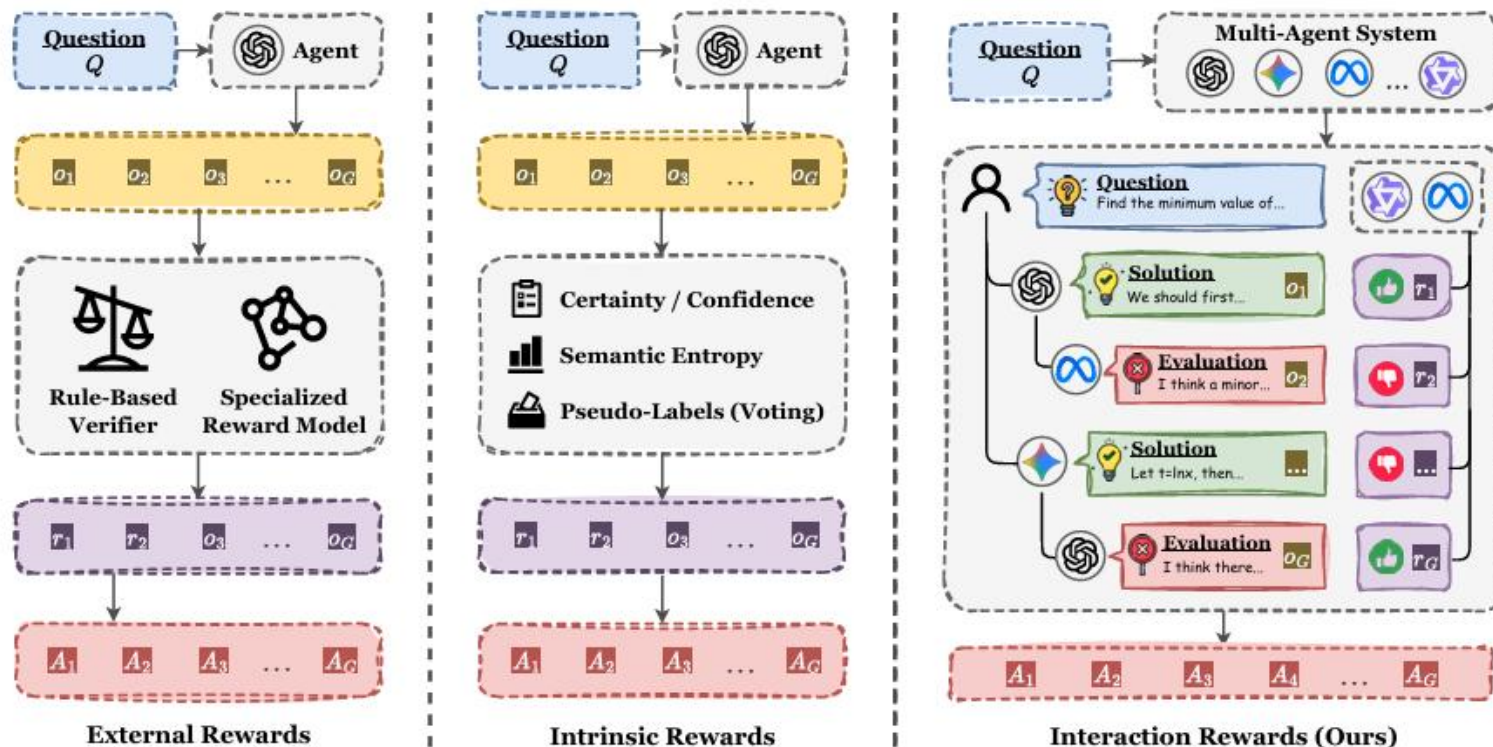


Figure 1: A comparison of our proposed CoMAS framework with existing RL-based self-evolution methods. The left column outlines methods utilizing external rewards from verifiers or reward models. The middle column outlines methods leveraging intrinsic rewards from metrics such as self-certainty, confidence, semantic entropy, and pseudo-labels from majority voting. The right column outlines our CoMAS framework, which derives rewards from multi-agent interactions.

Methods

Interaction → Reward Formulation → Policy Optimization

Interaction

系统里有 l 个agent:

$$U = \{u_1, u_2, \dots, u_l\}$$

Solution

agent 根据问题 q 和当前讨论历史 h_q

$$s_i = u_k(q, h_q)$$

Evaluation

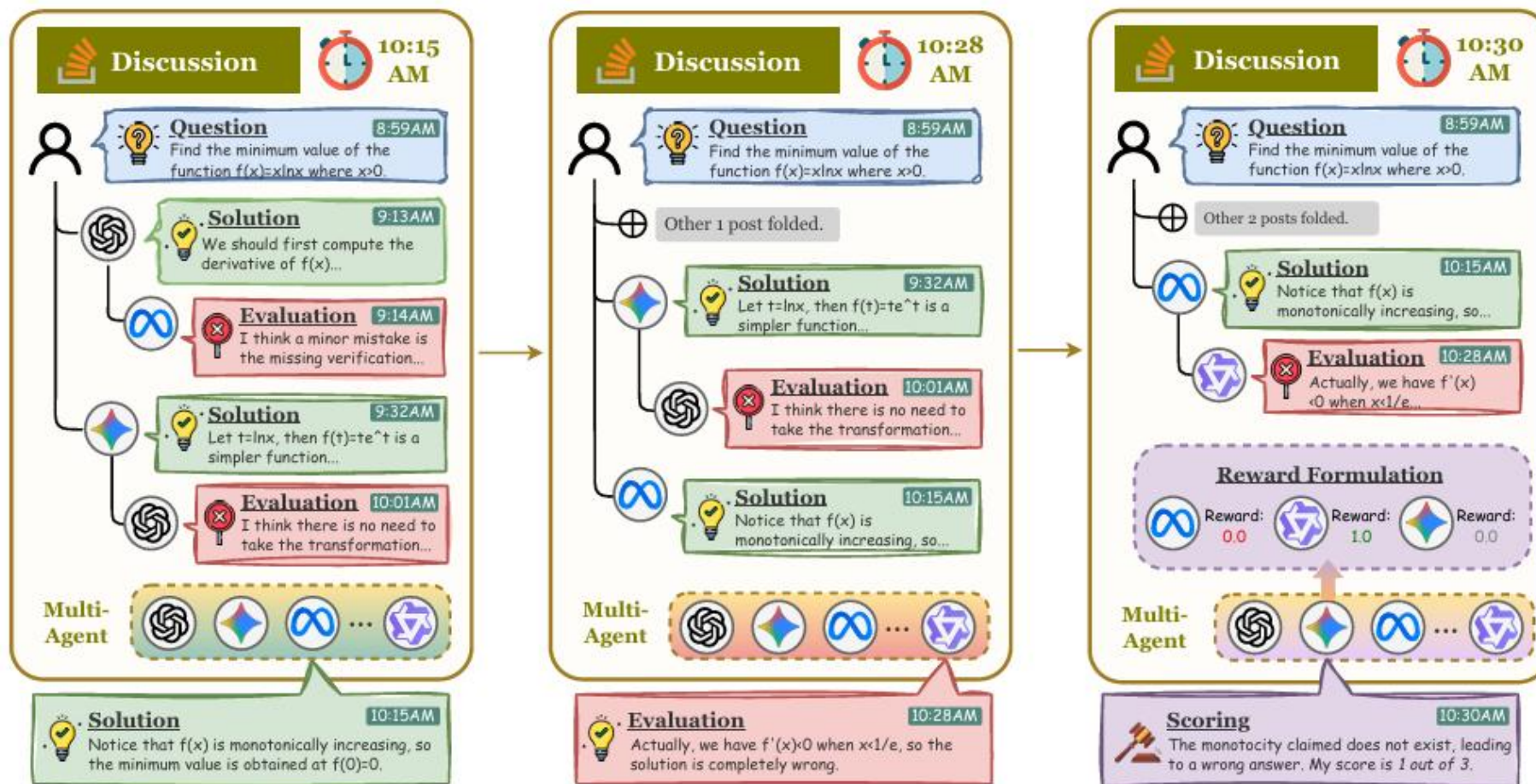
批判性评价

$$e_{i,j} = u_k(q, h_q, s_i)$$

Scoring

不加入历史 产生reward

$$\tau_{i,j} = u_k(q, s_i, e_{i,j})$$



Interaction → Reward Formulation → Policy Optimization

Reward Formulation

Score agent 输出1-3的整数:

Score 含义

- | | |
|---|------------------------------------|
| 3 | solution 正确, evaluation 没帮助或错误 |
| 2 | solution 基本正确, 但 evaluation 指出了小问题 |
| 1 | solution 错误, 并且 evaluation 找到了致命问题 |



给 solution 和 evaluation 分配 **互补的 reward**
solution reward + evaluation reward = 1

scoring agent 对输出格式给 reward

Policy Optimization

每个 agent 的交互样本都会被收集到自己的 replay buffer用于训练agent本身的policy

replay buffer

记录输入输出和reward $D_k = \{(p, o, r(o))\}$
且包含了solution evaluation scoring不同类型的样本

用REINFORCE++强化学习方法训练

训练数据是多角色、多上下文、多动作类型的
(context,output,reward)样本
使用KL散度限制模型更新幅度
提高高 reward 输出的生成概率
降低低 reward 输出的生成概率

Experiments



7种Benchmarks:数学、代码、科学回答

四种测试时的推理设置:

Vanilla: 单agent直接回答

Consistency: 多次采样后选择

AutoGen: 多agent协作框架

Debate: 多agent辩论框架

4种训练的baseline:

Untrained

SRLM: Self-Rewarding 机制

MAPoRL: debate框架中强化学习

TTRL: 多数投票生成伪标签

CoMAS: 在不同推理设置下更鲁棒

Table 1: The evaluation results of the agents trained by our framework on different benchmarks when employed in different setups, together with the comparison with the selected baselines. The results with performance improvements are highlighted in **green**, drops in **red**, and neutral changes in **gray**. CoMAS consistently improves the performance of the agents in almost all the settings.

		Method	GSM8K	MATH-500	HumanEval	Dataset MBPP	SciBench	GPQA	MMLU
Vanilla	Untrained		84.00	51.40	68.90	54.00	32.67	26.79	61.40
	SRLM		83.40 (-0.60)	52.20 (+0.80)	68.29 (-0.61)	53.80 (-0.20)	32.67 (+0.00)	27.01 (+0.22)	61.00 (-0.40)
	MAPoRL		84.80 (+0.80)	52.60 (+1.20)	69.51 (+0.61)	56.00 (+2.00)	34.07 (+1.40)	28.12 (+1.34)	61.40 (+0.00)
	TTRL		84.40 (+0.40)	53.40 (+2.00)	68.29 (-0.61)	57.40 (+3.40)	34.47 (+1.80)	25.45 (-1.34)	61.60 (+0.20)
	CoMAS (Ours)		85.40 (+1.40)	52.80 (+1.40)	70.73 (+1.83)	56.20 (+2.20)	34.67 (+2.00)	27.46 (+0.67)	62.40 (+1.00)
Consistency	Untrained		85.40	55.00	73.78	55.80	36.47	28.79	63.20
	SRLM		86.40 (+1.00)	55.40 (+0.40)	75.00 (+1.22)	56.20 (+0.40)	36.67 (+0.20)	29.24 (+0.45)	65.20 (+2.00)
	MAPoRL		85.80 (+0.40)	55.40 (+0.40)	75.61 (+1.83)	57.00 (+1.20)	39.08 (+2.61)	31.47 (+2.68)	63.20 (+0.00)
	TTRL		88.20 (+2.80)	56.80 (+1.80)	73.78 (+0.00)	59.00 (+3.20)	38.48 (+2.00)	27.23 (-1.56)	63.80 (+0.60)
	CoMAS (Ours)		87.20 (+1.80)	55.80 (+0.80)	77.44 (+3.66)	59.20 (+3.40)	37.68 (+1.20)	29.69 (+0.89)	65.60 (+2.40)
AutoGen	Untrained		52.60	38.40	39.63	29.80	20.24	16.29	37.40
	SRLM		58.00 (+5.40)	41.80 (+3.40)	44.51 (+4.88)	32.00 (+2.20)	21.24 (+1.00)	17.86 (+1.56)	42.40 (+5.00)
	MAPoRL		50.00 (-2.60)	37.40 (-1.00)	39.63 (+0.00)	34.60 (+4.80)	20.64 (+0.40)	21.65 (+5.36)	40.40 (+3.00)
	TTRL		41.00 (-11.60)	37.80 (-0.60)	23.17 (-16.46)	22.80 (-7.00)	19.64 (-0.60)	14.06 (-2.23)	34.00 (-3.40)
	CoMAS (Ours)		72.40 (+19.80)	45.80 (+7.40)	50.61 (+10.98)	38.00 (+8.20)	22.85 (+2.61)	22.99 (+6.70)	50.60 (+13.20)
Debate	Untrained		84.60	55.00	71.34	54.80	38.68	28.35	62.80
	SRLM		84.60 (+0.00)	54.80 (-0.20)	72.56 (+1.22)	53.60 (-1.20)	38.68 (+0.00)	28.57 (+0.22)	64.60 (+1.80)
	MAPoRL		85.40 (+0.80)	53.60 (-1.40)	74.39 (+3.05)	55.60 (+0.80)	39.88 (+1.20)	31.47 (+3.12)	64.80 (+2.00)
	TTRL		86.20 (+1.60)	55.20 (+0.20)	73.78 (+2.44)	58.00 (+3.20)	37.88 (-0.80)	29.02 (+0.67)	64.00 (+1.20)
	CoMAS (Ours)		85.20 (+0.60)	55.40 (+0.40)	77.44 (+6.10)	55.60 (+0.80)	39.08 (+0.40)	29.91 (+1.56)	65.20 (+2.40)

Scoring是否有必要?

去掉 scoring 后, agent 学会了利用 reward 机制, solver和evaluation相互支持, 从而拿到高 reward, 但并没有真正提升任务能力

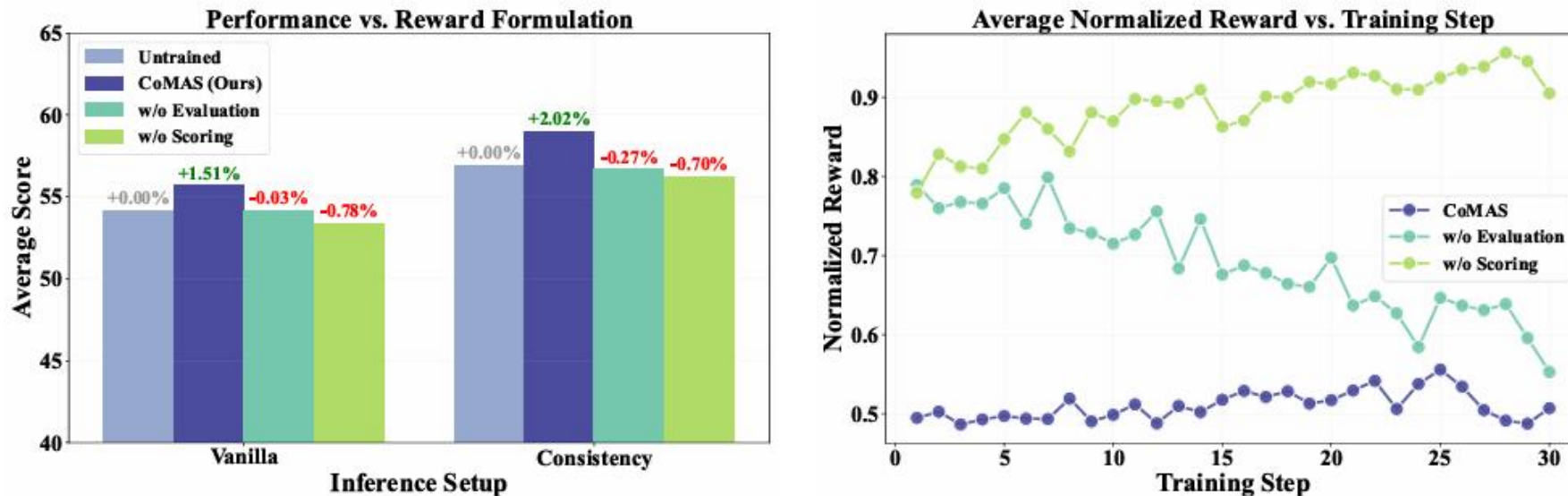


Figure 4: Results of the ablation study for reward formulation. The left figure compares the performance across the original CoMAS and two variants (without evaluation and without scoring). The right figure shows the average normalized rewards during the training process. These results indicate that the adversarial reward design is of key importance for the success of CoMAS.

Experiments

收益来自 inter-agent interaction, 更多 agents 可以带来更多样的 solution 和 evaluation

不同模型有不同的知识、推理路径和错误模式, 它们之间的互动更容易产生有价值的批评和判断

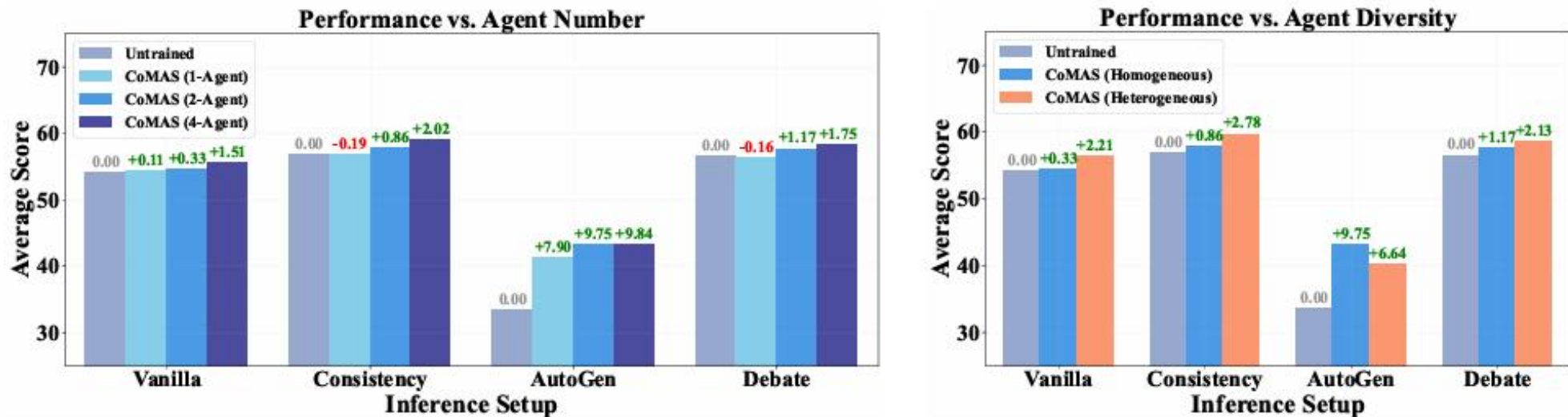


Figure 5: Results of the ablation study for framework scalability. The left figure shows how the number of agents affects the performance of CoMAS across different setups. The right figure compares the performance between homogeneous and heterogeneous agent settings. These results demonstrate the underlying scalability of CoMAS with the number and diversity of agents.

- Scorer是LLM-as-a-judge scorer判断失误会导致reward的错误分配
- Reward 设计比较粗糙
- 比较有意思的点：我觉得这种模式可能在开放性任务上更有潜力
 - 比如科研讨论、论文评审和复杂方案设计，往往没有明确标准答案，这正是 interaction reward 可能发挥作用的地方。多 agent 之间提出方案、批评方案、再由第三方判断质量，这种机制更接近人类协作学习。